The Articular Syste-

Definitions

The articular system is studied by arthrology.
An articulation, or joint, is the means of union between neighboring bony or cartilaginous parts.

Classification of joints

1-According to their structure:
Synarthroses: fibrous joints
Amphiarthroses: cartilaginous joints
Diarthroses: synovial joints

2-According to their mobility:
Synarthroses: Immobile
Amphiarthroses: Semi-mobile
Diarthroses: Mobile

I. Synarthroses :Fibrous joints / Immobile joints No articular cartilage.

a - Synfibroses: Found at the level of the skull and face.

b - Syndesmoses: The interosseous membrane between the two bones of the forearm or leg.

c - Gomphosis: A specific type uniting a tooth to a dental alveolus (socket) by fibrous tissue.

II. Amphiarthroses:

a - Synchondroses: Temporary joints during growth.
Located between the epiphysis and the diaphysis of long bones of the limbs: the growth plate (epiphyseal plate) responsible for the longitudinal growth of the bone.

b - Symphyses: These are semi-mobile joints.
The bony surfaces are united by a mixed fibrous and fibrocartilaginous structure reinforced by peripheral ligaments.
Example: The pubic symphysis.
The mobility of these joints is reduced.

III. Synovial Joints (Diarthroses):
Characterized by the existence of:

-An articular cavity.

-A synovial membrane.

-These are mobile joints.

-They constitute the majority of the limb joints.

Structure:

-Articular gap of only a few millimeters.

-Epiphyseal articular surfaces: of variable shape and covered with hyaline cartilage.

-An articular cavity.

-Means of union.

-Means of adaptation.

A. Constitution:

1. Articular surfaces:

The bony surfaces are covered with articular cartilage (hyaline cartilage), which is smooth and bluish-white.
The thickness of the hyaline cartilage is proportional to the pressure it supports.
This cartilage is not vascularized.
The degeneration of hyaline cartilage is responsible for arthrosis (osteoarthritis).

2. Means of union:

a. Articular capsule:

The capsule is a fibrous sleeve that inserts around the rim or vicinity of the bony articular surfaces. It comprises two parts:

-An external fibrous part: the fibrous membrane, of variable thickness and resistance.

-An internal part: the synovial membrane, which lines the inner face of the capsule.

The synovial membrane: is richly vascularized and innervated. It secretes synovia (or synovial fluid): a translucent, viscous fluid that facilitates the gliding of articular surfaces; it contributes to their nutrition.

b. Ligaments:

Are resistant means of union.
They are fibrous bands that unite the constituent parts of the joint.
They are the means of passive stabilization of a joint.
They are complemented by the tendons of periarticular muscles, which are means of active stabilization.

3-Means of Adaptation: These are fibrocartilaginous structures that exist in cases of articular incongruence and are interposed between the bony surfaces.
Three types of fibrocartilages: labrum, meniscus, and disc.

1. Menisci: Smooth fibrocartilaginous blades/crescents.
Role: Stability.
Examples: Knee (two menisci, one lateral and one medial) and the acromioclavicular joint.

2. The Labrum (or articular lip) : Fibrocartilaginous ring that inserts on the periphery of the bony articular zone.
They increase the depth and surface area of the articular zone.
Example: Scapulohumeral (shoulder) and coxofemoral (hip) joints.

3. Discs:Fibrocartilaginous partition.
Role: Shock absorption and distribution of pressure and traction constraints.
Example: Intervertebral disc and sternoclavicular joint.

Note:
These three types of fibrocartilages are devoid of vessels and nerves.
The absence of vascularization prevents the healing of a fibrocartilaginous tear.
Thus, a ruptured meniscal fragment causing clinical symptoms must be removed.

4-Bursae: Small pockets containing synovial fluid.
Role: Facilitates gliding and reduces friction between joints/structures.
Present at the level of the shoulder and knee.

B. Classification of Synovial Joints:
This classification is based on the morphology of the articular surfaces. We distinguish six types:

-Spheroid joint (Enarthrosis / Ball-and-socket).

-Ellipsoid joint (Condylar).

-Sellar joint (Saddle / Reciprocal reception).

-Trochlear joint (Ginglymus / Hinge).

-Trochoid joint (Pivot / Cylindrical).

-Plane joint (Arthrodia / Gliding).

1-Spheroid joint (Enarthrosis):
Unites two inverted spherical surfaces, one convex (full) and the other concave (hollow).
Examples: Scapulohumeral and coxofemoral joints.
They have great mobility, three axes of mobility.

2-Ellipsoid joint (Condylar)
Unites two ellipsoidal surfaces, one concave and the other convex.
Two axes of mobility.
Example: The radiocarpal joint (wrist).

3-Sellar joint (Saddle)
Or reciprocal reception; unites two bony surfaces that are convex in one direction and concave in the other.
Two axes of mobility.
Example: Carpometacarpal joint (thumb).

4-Trochlear joint (Hinge)
Unites two pulley-shaped surfaces, one full, the other hollow.
A single axis of mobility.
Example: The humero-ulnar joint at the elbow.

5-Trochoid joint (Pivot)
Cylindrical surfaces, full and hollow.
Examples: Proximal and distal radio-ulnar joints.

6-Plane joints (Arthrodia)

Unite two flat surfaces.
Example: Acromioclavicular joint.

IV-Articular Mechanics

1-Movements around a sagittal axis are:
-Abduction: moves a segment away from the median plane.
-Adduction: brings the segment closer to the median plane.

2-Movements around a transverse axis are:
-Flexion: "closes" the joint.
-Extension: "opens" the joint.

3-Movements around a vertical axis are:
-Lateral rotation: moves the anterior face of the segment outward.
-Medial rotation: moves the anterior face of the segment inward.

V-Exploration:

A. Clinical Examination:

The clinical examination of synovial joints explores the range and direction of movements in a bilateral and comparative manner. This examination is passive and active.

B. Imaging:

Standard Radiography (X-ray)
Arthrography
Ultrasound and especially MRI
Arthroscopy:

C. Pathology:

Non-traumatic pathologies: arthritis (infectious or inflammatory) and degenerative arthrosis (osteoarthritis).
Traumatic articular pathologies: the most frequent being luxations (dislocations - loss of contact between two articular surfaces) and entorses (sprains - ligament rupture or elongation).